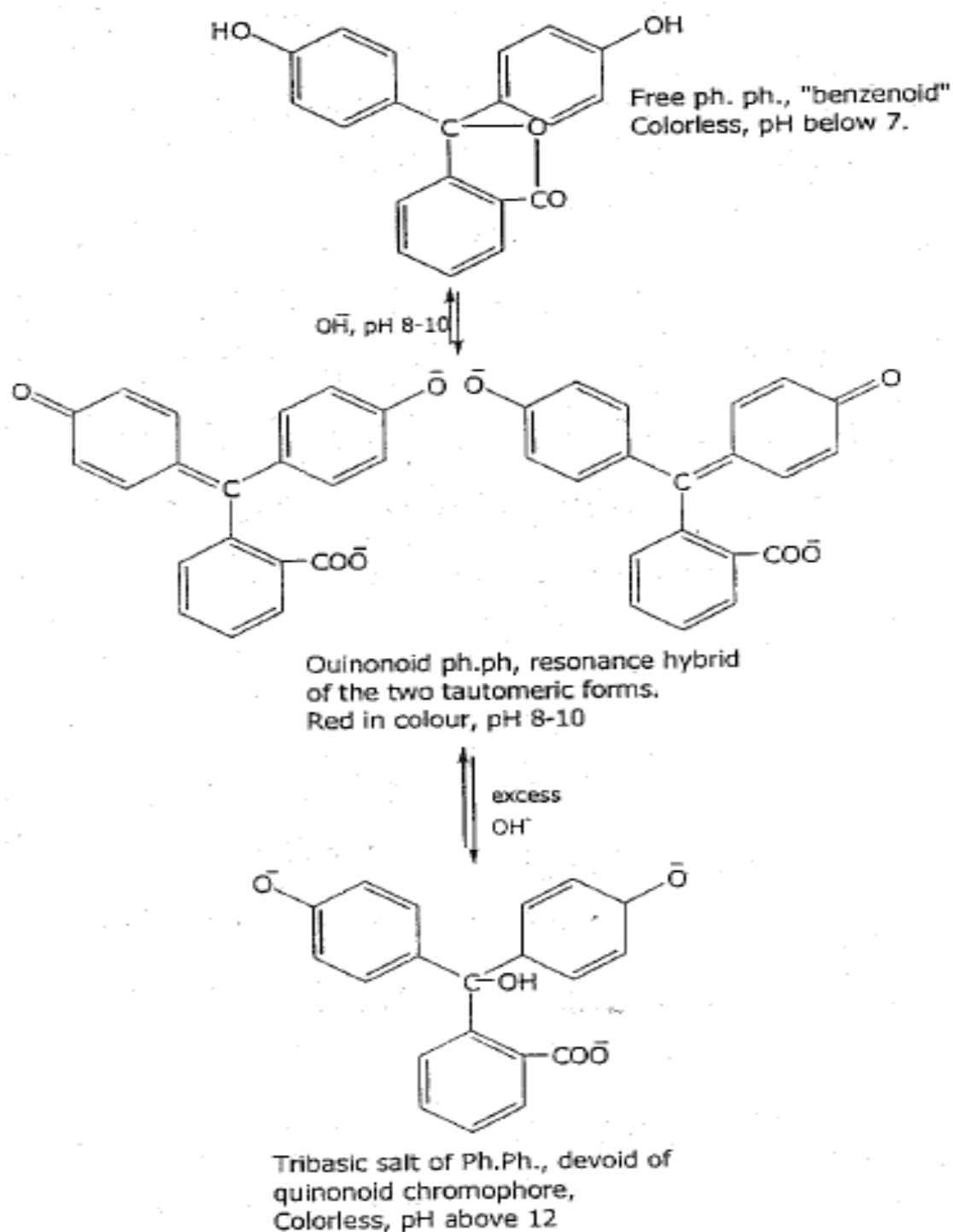
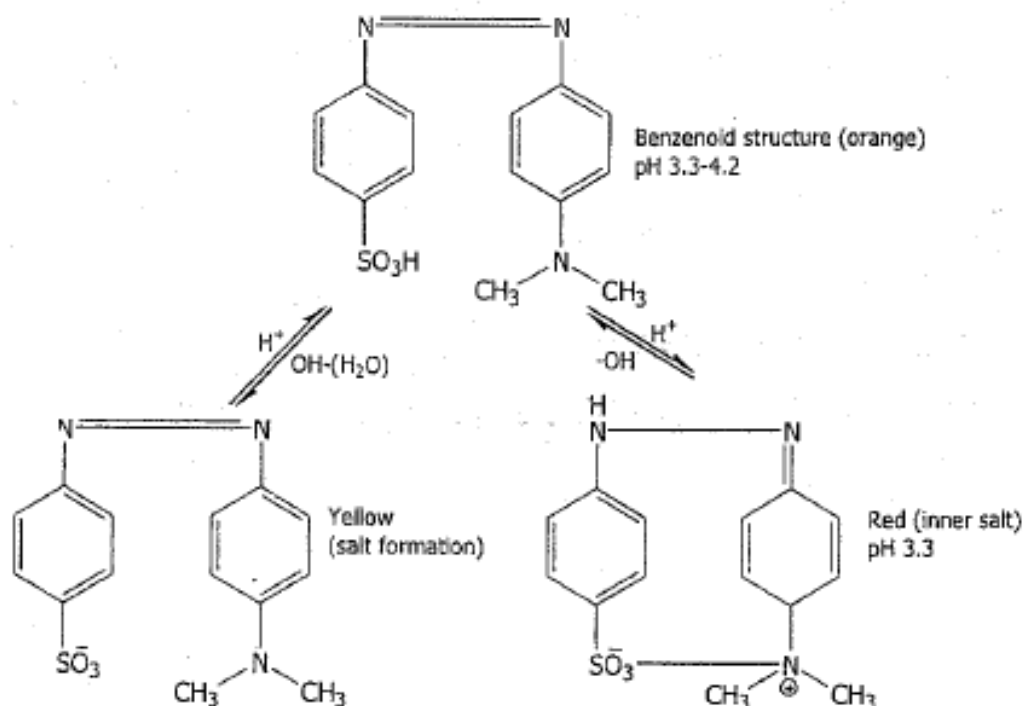


- COLOR CHANGE OF SOME COMMON INDICATORS

- PHENOLPHTHALEIN (ph.ph.)



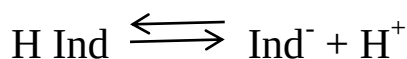
METHYL ORANGE (M.O)



Indicator Constant, Effective Range of Indicators and Transition Interval.

Although the change of indicator color is due to molecular rearrangement, the equilibrium according to Ostwald is useful in illustrating the mass action effect of the addition of acid and base.

For an acid indicator:



- Unionized, (acidic color) ionized, (basic color)
- $[\text{H}^+][\text{Ind}^-]/[\text{H Ind}] = K_{\text{ind}}$ "indicator constant)
- $[\text{H}^+] = K_{\text{ind}} [\text{H Ind}]/[\text{Ind}^-]$

$$\text{pH} = \text{p}K_{\text{ind}} + \log [\text{Ind}^-]/[\text{Hind}]$$

$$\text{pH} = \text{pK}_{\text{ind}} + \log [\text{basic color}]/[\text{acidic color}]$$

When the concentration of the basic color is equal to that of acidic color, then;

$$\text{pH} = \text{pK}_{\text{ind}}$$

This is sometime known as the "middle tint" of the indicator, and applies only if the two colors are of equal intensity.

If one form is more intensely colored than the other or the eye is more sensitive to one color than the other, then the change from one color to the other can only be detected when there is a slight excess of the produced color. The color change of an indicator, therefore, depends on the ratio $[\text{basic color}]/[\text{acidic color}]$, which should not exceed 1/10 or 10/1, so that :

$$\text{pH} = \text{pK}_{\text{ind}} \pm 1$$

This equation determines the effective range of an indicator.

Within this range the color change will be gradual, the interval during which shades of color are produced extending between the two-colored forms is termed the transition interval. The effective range of an indicator is the pH units over which it changes its color, for a good indicator this should not exceed two pH units, e.g. phenolphthalein: (8-10), methyl orange: (3.3-4.4) and methyl red (4-6). The color change interval of most of the various indicators is represented graphically in figure 1.

The change of the indicator from one form to another actually uses up or liberates some hydrogen ions.

Ideally this quantity of hydrogen ions must be negligible in comparison to the acid involved in the main titration reaction, otherwise a significant error is introduced. The magnitude of such an indicator error increases as the concentration of indicator increases. For this, the indicator concentration should not be any larger than to render, the end point change visible.

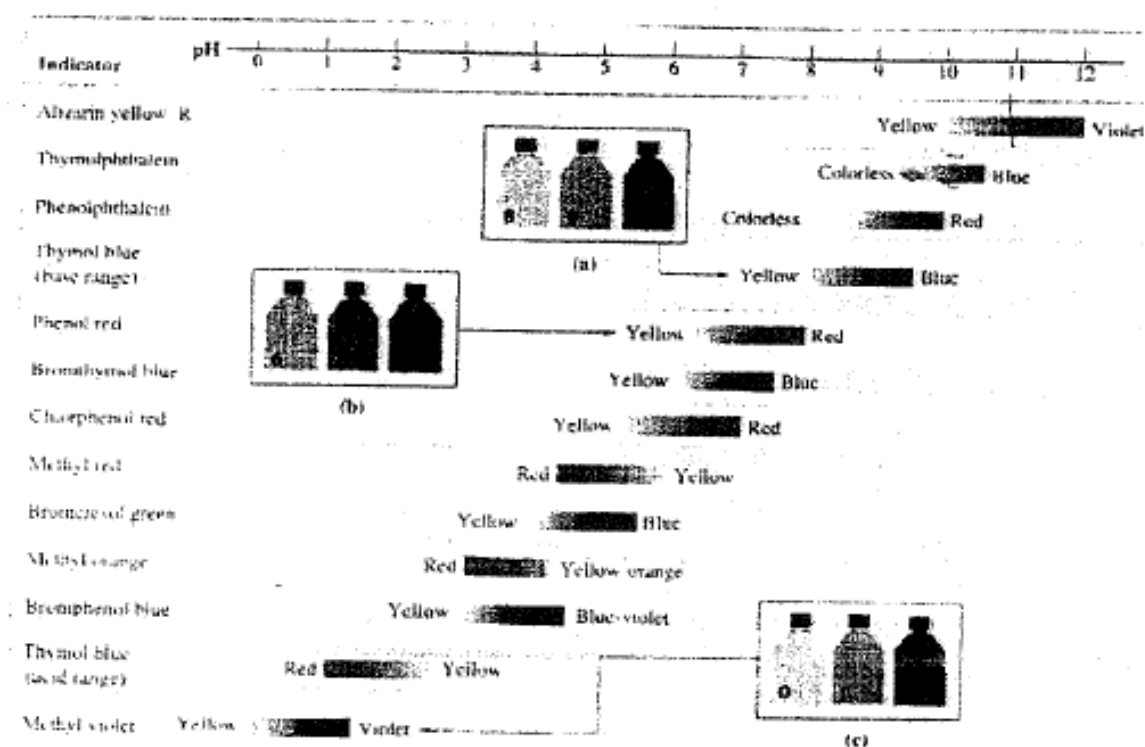


Fig. 1: Color change intervals of most common indicators.

A- Screened Indicator

With some indicators, e.g., methyl orange, the color change is not easily detectable particularly in light. A sharper and more pronounced color change may often be obtained by using a suitable mixture of the indicator and a dye, e.g.,

methyl orange with indigocarmine. The color change, at pH 4, with such a screened indicator is from yellowish green "alkaline" to violet "acid"

B- Mixed Indicators

A sharper color change may be obtained by using a mixture of two indicators having similar pH but showing contrasting color, e.g. thymol blue and cresol red.

c- Universal or Multirange indicators

By suitably mixing certain indicators, the color change may be made to extend over a considerable pH range. Such mixtures are usually called universal indicators. They are not suitable for quantitative titration but may be used for the rough determination of pH of solutions.

Note: color indicator cannot be used if colored solutions are to be titrated. Equivalence point in such cases can be detected either potentiometrically or by the use of turbidity or fluorescence indicators.

2- TURBIDITY INDICATORS

Many higher organic acids or bases form colloidal solutions which, at certain pH, form flocculent precipitates. Their use is limited only to case when the use of color indicator is not practical, e.g. when titrating a dark colored solution.

3-FLUORESCENCE INDICATORS

Certain compounds emit visible radiation when exposed to ultraviolet light. This property may stop, or intensify, when certain pH is reached. Fluorescence indicators, e.g. umbelliferone, are used to detect end point when color or turbid solutions are titrated.

It should be noted that indicators are only used to indicate a narrow of pH which might be in the alkaline, neutral or acidic side. Water with a pH 7, is alkaline to methyl orange "range 3.3- 4.4" and is acidic to phenolphthalein "range 8-10". It is only neutral to indicators with a range between 6-8 such as litmus, cresol red and bromothymol blue.

In neutralization reactions, the reaction is completed when the acid or base to be titrated to the salt. The pH of the salt solution at the expected concentration of the titration has to be calculated using the equation given before. **Consequently, the suitable indicator, with a pH range coinciding with that of the salt, is chosen.**

NEUTRALISATION TITRATION CURVES

For neutralization reaction titration, the titration curve is a plot of pH versus the volume of titrant. Titration curves vary in character according to the reactants and their concentrations.

Titration curves are constructed to study the feasibility of the titration, and to help in choosing an indicator for the

titration. In a titration curve we have to calculate points corresponding to different volume of titrant, and to calculate the pH after each addition. Points must be calculated: **before** titration, **during** titration, at the **equivalence point**, and **beyond** the equivalence point. The equations, derived previously, are to be used to construct such titration curves. The pH values during the titration process may be also measured experimentally. **In** the following examples, the mathematical calculation of the pH, during the course of titration has been applied.

1- Strong Acid-Strong Base Titration

e.g., HCl against NaOH

the equation used to calculate pH before and during titration is:

$$\text{pH} = \text{pC}_a$$

While, at the equivalence point:

$$\text{pH} = \text{pOH} = \frac{1}{2}\text{pK}_w = 7.0 \text{ (neutral)}$$

While, at the equivalence point, the excess NaOH added controls the pH of the medium, thus the pH is calculated as follows.

$$\text{pH} = \text{pK}_w - \text{pC}_b$$

Figure 2 shows the titration curves 20 ml of 0.5N HCl against

0.5 N NaOH. The titration curve is influenced by concentration of the titrating agent, the more dilute

the acid or base used in the titration, the less abrupt is the inflection at the equivalence point.

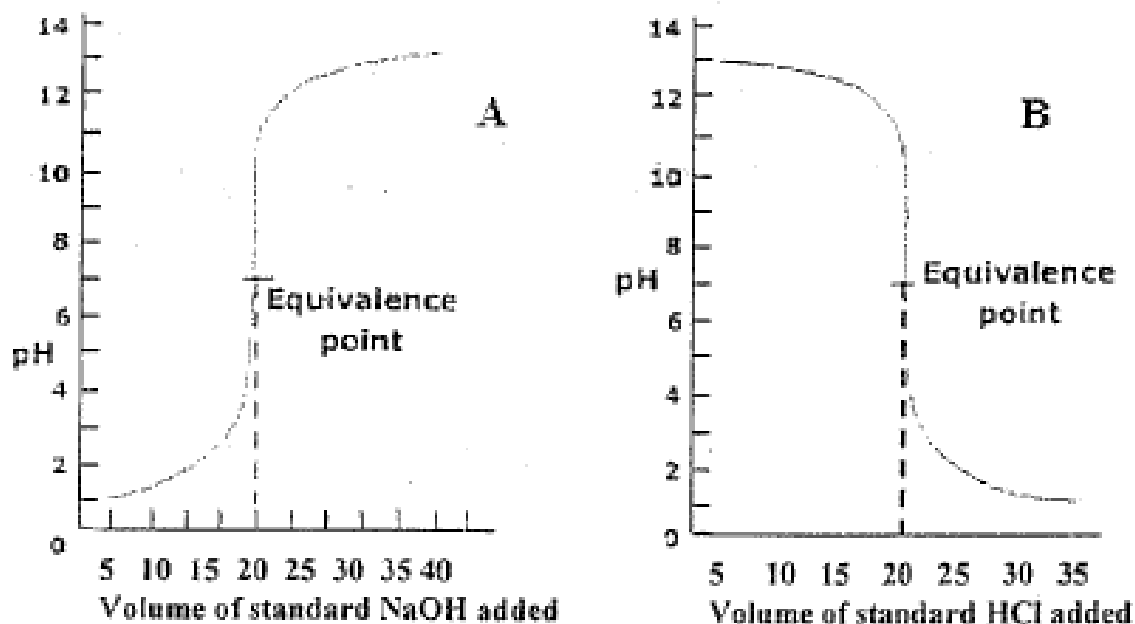


Fig. 2: Neutralization curves of standards HCl and NaOH.

(A) Titration of a strong acid with a strong base & (B) titration of strong base with a strong acid.

From the curves in figure 2, it is obvious that when titrating a strong acid against a strong base the pH of the solution rises very slowly until 99.9% of the acid has been neutralized. By adding the remaining 0.1 ml of alkali, the pH sudden rises to 7. Exceeding the 100% neutrality point by as little as 0.1 ml alkali; the medium becomes alkaline and again the pH rapidly rises to about 10. Further addition of alkali gradually rises the pH until when 200% alkali has been added, pH 14 is reached. Near the neutrality point, i.e.

between 99 and 101%, and when compared with color change intervals of various indicators, the proper one can be chosen, which will show exactly the neutrality point. Actually, as can be seen from the figure, the change in the pH is so rapid at and near the equivalence point that any indicator with a color change interval between pH 3.5 and 10.5 could be used without much error.

2- Weak Acid-Strong base Titration, eg. CH_3COOH and NaOH .

Let us consider the titration of 100 ml of 0.1 N acetic acid with 0.1 N sodium hydroxide solution.

The initial pH of 0.1 N acetic acid [$\text{pK}_a = 4.74$] is calculated from the equation

$$\begin{aligned}\text{pH} &= \frac{1}{2}\text{pK}_a + \frac{1}{2}\text{pC}_a \\ &= 2.37 + 0.5 = 2.87\end{aligned}$$

When 50 ml of 0.1 N alkali has been added, the pH is calculated from the formula for buffers:

$$\begin{aligned}\text{pH} &= \text{pK}_a + \log \frac{[\text{Salt}]}{[\text{Acid}]} \\ &= 4.74 + \log 50/50 = 4.74\end{aligned}$$

and when 90 ml of alkali has been added then

$$\text{pH} = 4.74 + \log 90/10 = 5.7$$

The pH of the solution at the equivalence point is given by the equation:

$$\text{pH} = \frac{1}{2} \text{pK}_w + \frac{1}{2} \text{pK}_a - \frac{1}{2} \text{pC}_s$$

Note that pC_s at the equivalent =

$$-\log (100 \times 0.1/200) = 1.3$$

$$\therefore \text{pH} = 7 + 2.7 - 0.65 = 8.72$$

After the equivalence point has been passed, the solution now contains excess OH^- , the pH may be assumed "with sufficient accuracy" to be that to excess of strong base present. Results are illustrated in figure 3. It is evident from that figure that neither methyl orange nor methyl red can be used as indicators for O.S.N acid. The equivalence point is at 8.7 and it is necessary to use an indicator with a pH-range on the slightly alkaline side, such as phenolphthalein [pH-range: 8.3-10] or thymolphthalein [pH-range: 9.3-10.5].

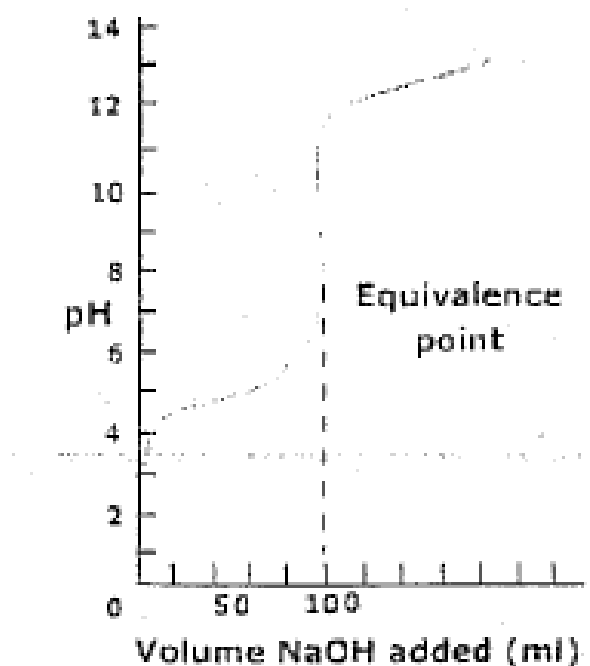


Fig. 3: Neutralization curve of acetic acid, sodium hydroxide

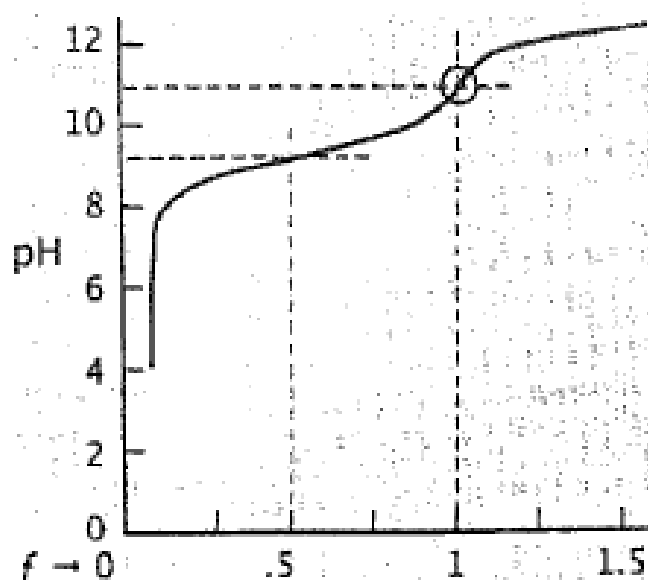


Fig. 4: Titration of very weak acid (boric acid), with a strong base (0.1N NaOH)

The titration of acid with ionization constant less than about 10^{-7} , like boric acid (H_3BO_3) is not feasible. In such a case, the inflection at the equivalence is not sharp, as the change of color is gradual and extends over a considerable volume of alkali.

Figure 4 show the titration curve of 0.1 N boric acid with 0.1 N sodium hydroxide after dilution to 100 ml. The ionization constant of H_3BO_3 ($H_3BO_3 = H^+ + H_2BO_3^-$) is 5.5×10^{-10}

3- Weak Base-Strong Acid Titration e.g. NH_3 and-HCl

Suppose the titration of 100 ml 0.1 N ammonia $pK_b = 4.74$ "with 0.1 N hydrochloric acid solution. The equations used at the various stages are as follows:

At the start the pH is calculated from the formula weak bases:

$$pH = 14 - \frac{1}{2} pK_b - \frac{1}{2} pC_b$$

$$14 - 2.37 - 0.5 = 11.13$$

During the titration, the pH is calculated using the formula for buffers e.g. when 90 ml 0.1N HCl is added:

$$pH = 14 - pK_b + \log \frac{[Base]}{[Salt]}$$

$$14 - 4.74 + \log \frac{10}{90} = 8.31$$

At the equivalence point the pH is given by the equation of salts:

$$pH = \frac{1}{2} pK_w - \frac{1}{2} pK_b + \frac{1}{2} pC_s$$

$$= 7 - 2.37 + 0.65 = 5.13$$

After the equivalence point, the solution represents a strong acid solution and its pH can be calculated in the usual manner as illustrated in figure 5.

It is clear that neither thymolphthalein nor phenolphthalein can be used in the titration of 0.1N aqueous ammonia. The equivalence point is at pH 5.1, and it is necessary to use an indicator with a pH range on the slightly acid-side "3-6.5", such as methyl orange or methyl red. For the weak base, $K_b = 1 \times 10^{-7}$, methyl orange may be used. No sharp color change will be obtained with methyl red, and the titration error will be considerable.

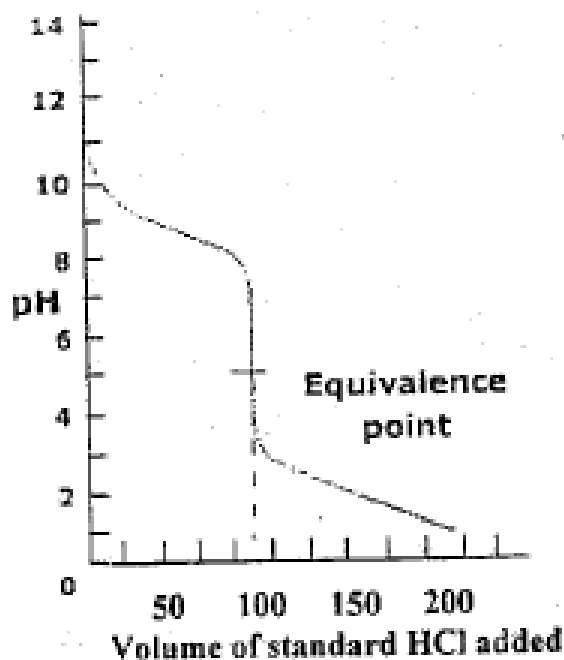


Fig. 5: Neutralization curve of aqueous ammonia,

($K_b = 1.8 \times 10^{-5}$) with 0.1 N hydrochloric acid.

4- Weak Add-weak base Titration, e.g. NH_3 and CH_3COOH

The titration curve in such cases is smooth and the change of pH at the vicinity of the equivalence point is very small. Such titration is not feasible and should be avoided.

SOME APPLICATIONS OF NEUTRALIZATION REACTIONS

Acidimetry is the process of determining acidic substances by titration, and alkalimetry is the corresponding process for determining alkaline substances, and both processes are neutralization processes. A detailed description of determinations of various acids, bases and even neutral salt compounds, a variety of procedures, are summarized as follows: